

EXAM 7 – SPRING 2014

6. (1 point)

Given the following information from an insurance company's portfolio of policies with a \$500,000 deductible:

Age-to-Age Loss and ALAE Development Factors

	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>
Unlimited	1.912	1.316	X	1.077
Limited to \$500,000	1.900	1.300	1.125	1.050

Limited Severity Relativities (R^L)

<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>
0.980	0.974	0.962	0.944	0.920

a. (0.5 point)

Calculate the missing value, X, from the table above.

b. (0.5 point)

Calculate the excess of \$500,000 36-48 month development factor.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 6	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: A4: Estimate unpaid claims for various layers of claims.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> $\text{LDF}_t^L = \text{LDF}_t \times R_{t+1} / R_t$ $1.125 = \text{LDF}_t \times (.944/.962)$ $\text{LDF}_t = 1.146$ <i>Sample Answer 2</i> $1.125 = X \times .944/962$ $X = 1.146$	
Part b: 0.5 point	
<i>Sample Answer 1</i> $\text{LDF}_t = \text{LDF}_t^L \times R_t + (1-R_t) \times \text{XSLDF}$ $1.146 = 1.125 \times .962 + (1-.962) \times \text{XSLDF}$ $\text{XSLDF} = 1.690$ <i>Sample Answer 2</i> $1.146 \times (1-.944)/(1-.962) = 1.6895$	
EXAMINER'S REPORT	
Candidates generally scored well, with almost 75% receiving full credit of 1.0 and less than 10% received 0.25 point or less. On average, candidates had similar success with a. and b. The most common mistake was selecting an incorrect limited severity relativity, be it simple column reference error or just misunderstanding the formula/concept.	
Part a	
The candidate was expected to provide a correct solution, but not via one and only one formula. The most common mistake was selecting an incorrect limited severity relativity, be it simple column reference error or just misunderstanding the formula/concept.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Part b
The candidate was expected to provide a correct solution, but not via one and only one formula. The most common mistake was using an incorrect age for the LDF/relativity relationship.